



UNIVERSITÄT ZU LÜBECK

Trustworthy & Human-Centered AI

Project: Study Design & Report

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Steps

1. **Find a group** (2-3 people)
2. **Choose a topic** discussed during the lecture
3. Define your concrete **research question**
4. Come up with a **study plan** how to test your research question

--- at this point you should upload a short video (max. 5min) explaining your concept ---

5. **Implement** your idea
6. **Test some participants** (e.g. other students, friends)
7. **Analyse your data** (be careful with interpretations due to probably small sample sizes!)

--- hand in a project report ---

Rough (Preliminary) Schedule

- **As soon as possible (end of April?):** Find a group (2-3 people)

Note: We expect you to find a group by yourself. If you tried but still have problems, get in contact with us. Please do so early enough. Once the project started, it might be difficult to integrate in an already ongoing group work.

- **End of April:** Ideas, framework and rough requirements of the project will be presented
- **Beginning of May:** Technical details and information regarding the study design will be presented

--- develop a concept, research question, ideas for the implementation, test the environment ---

- **End of May / beginning of June:** Upload a short video explaining your concept

Note: These videos will be uploaded in Moodle and we expect to watch the videos of the other groups to also compare and discuss different approaches in your final report. We will also give some feedback and might open feedback options from other students for further discussions.

--- implementation, data collection, analysis ---

- **Mid of July:** Hand in the final report

Introduction & Background

- **Motivation and relevance** of the topic (usually a good start for your paper to arouse interest)
- **Theoretical background:** You should place your study in a theoretical background from
 1. the lecture (and material that is referred to) and
 2. other literature such as studies or other papers.
- **Research gap:** It should be clear where the **research gap** is and how your **research question** addresses it. Additionally, your **hypothesis** should be concrete and stated clearly.
- **Definition of terms:** Please define the terms you use. One example is trustworthiness. In the course of this semester, you will get to know various aspects from different perspectives. You should make clear what you mean by it and what exactly you are referring to, e.g. which aspects you are focussing on.
- **Brief outline** of the paper's structure
- **[Ethical approval:** Usually you would need to apply for an ethical approval before collecting participant data. Since this is only a small (let's call it pilot) study, this is not necessary.]

Examples of Research Questions

- Is there a change in the perceived trustworthiness of a system after changing [some component of] the **explanation** or adding **feedback** to the task?
- Is it possible to increase the sense of well-being when working with a system by adding **notifications** (e.g. warning or motivation messages)?
- How do **mistakes / false alarms / an increased frequency of interim notifications** by the system influence the likelihood that individuals will choose to work with the system again at a later date?
- Do changes in the **interface** affect how fast people respond to alarms?
- Do participants who have a **time pressure** while performing the task make more mistakes than participants without a time pressure?
- Does a **time delay** [of messages / warnings / system's responses] cause increased frustration among participants?
- How does the (perceived) **cognitive load** (e.g. through adding different components of the task or adapting the difficulty) affect the perceived reliability of a system?

These are just some rough ideas. You should have a look at the system yourself to check what is possible!

Examples of Hypothesis

- The perceived trustworthiness of the system differs between participants under time pressure and those without time pressure.
- Increasing the time delay of system responses beyond two seconds will lead to higher levels of frustration among participants.
- Participants with higher prior expectations regarding the utility of the system will perceive false messages as more disruptive than participants with lower prior expectations.

... could for example be (statistically speaking) a correlation of two continuous variables or a comparison of two groups with different conditions

Note:

1. Your hypothesis should be falsifiable !!
2. Don't make it too complex. Keep in mind that you should be able to...
 - (a) implement it in the system and
 - (b) perform the data analysis.

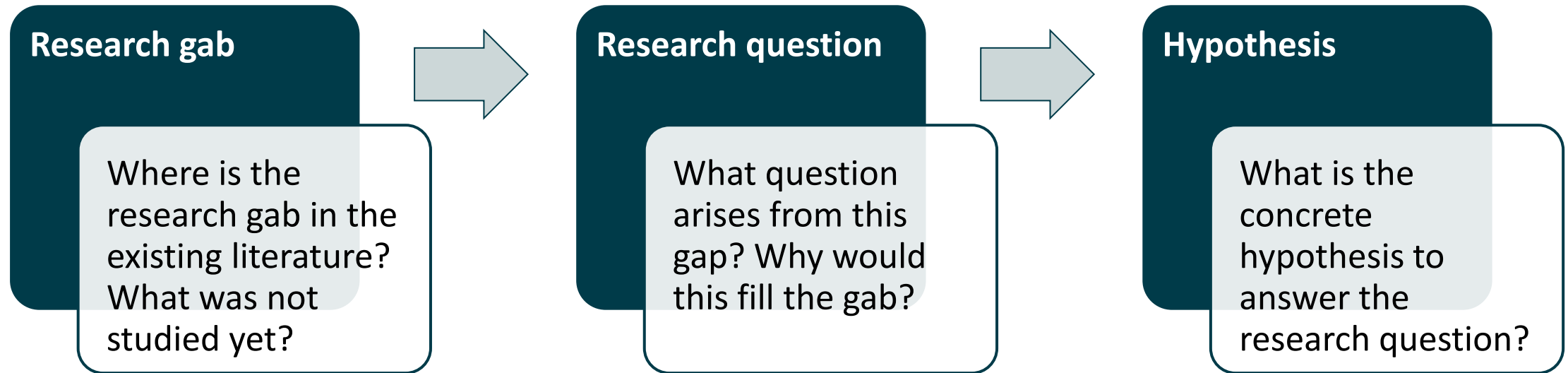
Example **NON**-falsifiable hypothesis

“A system is trustworthy because it is inherently good.”  please **DON'T** do that! ;)

Why is it not falsifiable?

- The phrase "inherently good" is vague and subjective.
- There are no measurable or testable criteria to evaluate the claim.
- Even if the system fails or causes harm, someone might argue: "Those were just exceptions; the system is still inherently good."

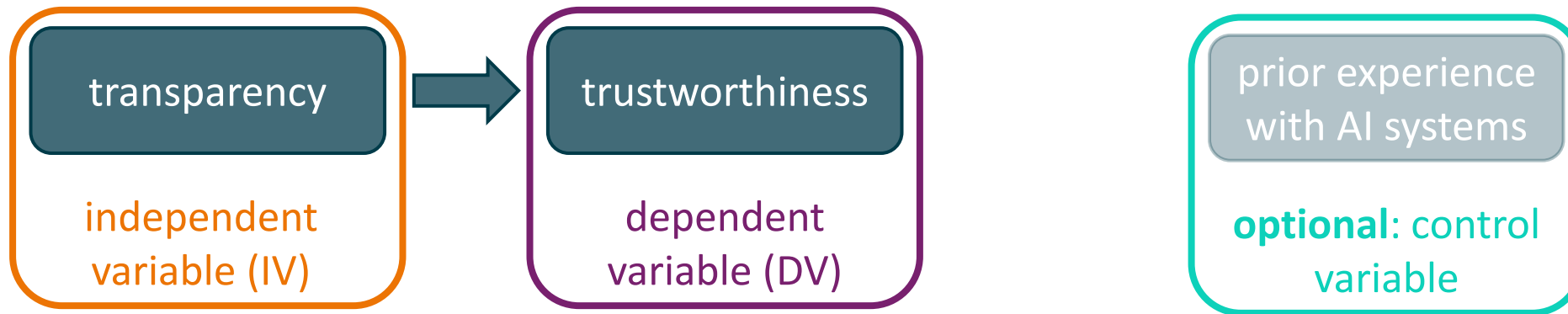
Connection of RG, RQ, and Hypothesis



- You should link these three aspects, especially the research question and your hypothesis.
- It should be clear which hypothesis belongs to which RQ (especially if you have more than one).
- You can for example list the research question and underneath the hypothesis.
- It is a good idea to number the hypotheses so that they can be referred to more easily later on.

Variables (One Example)

- Hypothesis: “A more transparent communication increases the perceived trustworthiness of X.”



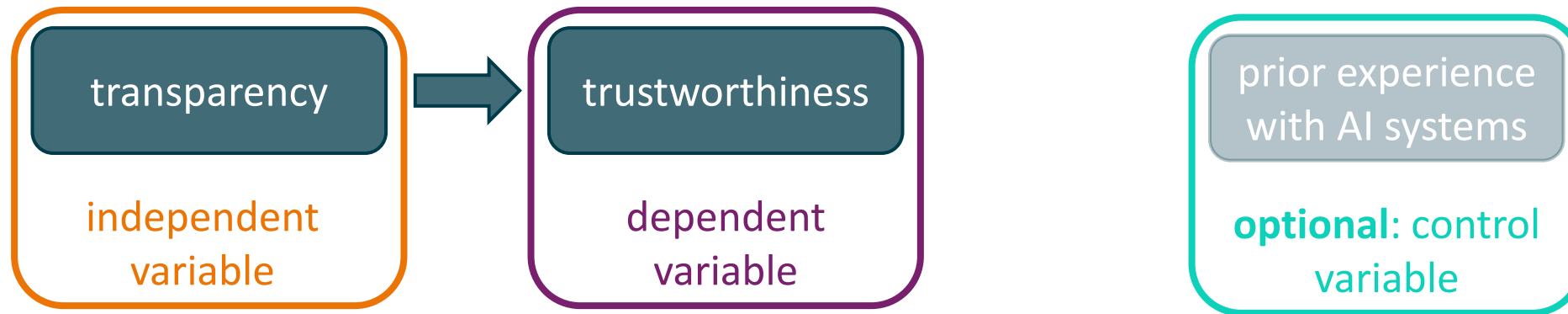
Operationalisation



- **IV:** What exactly do I have to change in a system in order to include a factor (e.g. time pressure from a ticking noise compared to an elapsing timer)?
- **DV:** What can I use to measure a certain construct (e.g. using a questionnaire, reaction time)?
- You should make clear why you operationalised your variables in such way!

Variables (One Example)

- Hypothesis: “A more transparent communication increases the perceived trustworthiness of X.”



Operationalisation

- **IV:** Two groups
 1. low transparency: system gives only final decision
 2. high transparency: system provides decision plus explanatory rationale
- **DV:** For example an (existing) trustworthiness questionnaire with responses ranging from 1 to 5

Don't Mix Your Conditions

Hypothesis: Timing × Clarity: High message clarity reduces the negative impact of late notifications.

High-clarity condition: a short and clear warning message (2 sentences) is displayed after 2 seconds

Low-clarity condition: a long and vague warning message (1 word) is displayed after 5 seconds

Problem: Not only the clarity but also the timing and length of the message varies within the clarity condition. Therefore, it is hard to say what is causing your effect (clear vs. long vs. earlier message).

Don't mix! Timing in clarity conditions should be similar as well as clarity in timing conditions.

Better example: approximately equal length and number of words

- Clarity: Concrete (e.g. "pressure is too high") vs. vague ("something is wrong")
- Timing: Immediate (e.g. after 0 second) vs. delayed (e.g. after 5 seconds)

Don't Mix Your Conditions

Hypothesis: Timing × Clarity: High message clarity reduces the negative impact of late notifications.

Better example:

- Clarity: Concrete (e.g. “pressure is too high”) vs. vague (“something is wrong”)
- Timing: Immediate (e.g. after 0 second) vs. delayed (e.g. after 5 seconds)

Groups / Conditions:

Timing / Clarity	Concrete	Vague
Early	Group / Condition 1	Group / Condition 2
Delayed	Group / Condition 3	Group / Condition 4

Operationalise Trustworthiness

- Before you choose a question(naire), you should first think about the concrete concept.
- You should ask yourself what exactly you would like to measure.
- Keep in mind that trustworthiness is a broad concept and includes different aspects.
- Your results might differ depending on how exactly you ask or what aspect(s) you choose.

You can...

- Search for an existing questionnaire (there are many) – this is recommended!
- Develop questions yourself (can be tricky!)
 - You should **NOT** just ask “Do you think this system is trustworthy?” – be specific!
 - How should the participants respond? Open answer, yes/no, Likert Scale (e.g. 1-5 or 0-7)

... It is generally not good practice to develop questions yourself (without evaluating them). If you do so nonetheless – as an exception in the context of this small study – you need to clearly explain why you chose these specific questions and why existing questionnaires did not meet your objectives.

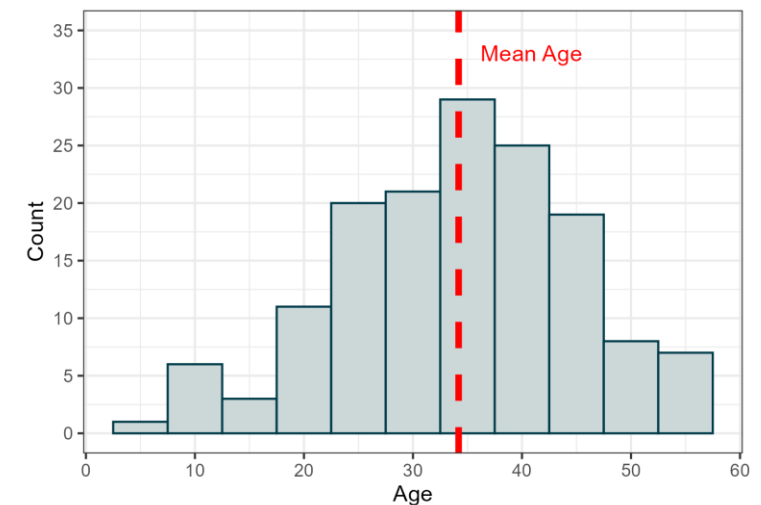
Study Design and Participants

Within-Subject Design	Between-Subject Design
Repeated Measures	Independent Measures
<i>Same</i> participants for <i>all</i> conditions	<i>Different</i> participants for each condition
Usually less participants needed (better statistical power)	Usually more participants needed (more groups and more noise)

⇒ Maybe: Randomise order of conditions

Sample

- Sample size probably won't be sufficient for a real conclusion and not representative – fine here
- Good enough (for us): Approx. 10 participants (or 5-6 per group), e.g. family, friends, colleagues
- Please collect and report demographic data (age, gender)



Study Design and Participants

- **Within-Subject** (Repeated Measures): The *same* participants are exposed to *all* conditions.
- **Between-Subject** (Independent Measures): *Different* participants are assigned to each condition.
- You usually need **more participants in a between-subject design** (compared to within-subject) to have the same (statistical) power because (1) you need enough participants per groups and (2) each participant adds additional noise (whereas they are their own control in a repeated measures design).
- Don't worry, we know that it won't be a sufficient sample size and that it probably **won't be representative** (if you only ask people in your immediate environment, for example) - so everything is **much less extensive** than for a real study.
- In total **approx. 10 participants** would be great (everyone in a group can ask some people). If you have different groups in a between-subject design, **5-6 participants per group** are okay.
- If you have a within-subject design, please consider **order effects**, such as learning over time. You might want to think about **randomising** the order of conditions for each participant.
- Please collect and report **demographic data** of the participants (e.g. age, gender).

Briefing & Debriefing

- **Briefing:** *Inform participants* about what they are about to do, in clear and understandable terms.
- **Debriefing:** *Explain the true nature* of the study once participation is over.
- You should report information about both in the report!

What **NOT** to say during the briefing?

- Your concrete research question (depending on whether it could influence the participants)
- Your hypothesis
- Your expectations about the outcome
- ... everything that might affect the behaviour of the participants – to avoid a bias !

You should of course explain what it was about *afterwards* in the debriefing.

Briefing: Good and Bad Example

Bad briefing example: "Welcome! In this study, we are testing whether people trust AI chatbots more when they give detailed explanations. We expect that you will find the AI that explains itself much more trustworthy. Please answer the survey questions honestly to help confirm this."

Why?

- Reveals the hypothesis
- Pressure the participants
- Induces bias
- Potentially invalid results

Good briefing example: "Thank you for joining this study on human interaction with AI systems. You will interact with different AI models and then answer questions about your experience. Your participation is voluntary, and you may stop at any time without penalty. All your responses are confidential and will only be used for research purposes."

Data Analysis

Statistical analysis (incl. test for the assumptions!), e.g.

- t-Test: group comparison (within vs. between subject design)
- Correlation or (linear) regression (for continuous variables)
- Nonparametric or Bayesian approaches

Please **only use statistical tests if you are familiar with them!** Otherwise: please don't!

Visual evaluation, e.g.

- Comparison (of categorical variables): e.g. bar chart, dot plot
- Distributions (of continuous variables): e.g. histogram, boxplot
- Relationships (at least two variables): e.g. scatter plot, line graph

Descriptive statistics: Mean, median, variance, standard deviation, ...

Results

- Objective presentation of findings (no interpretation yet)
- Report key patterns and numerical outcomes, use tables, figures, and statistics where relevant
- State what is used for data analysis (and cite correctly) – e.g. R Studio, Python, Matlab, ...
- If you used statistical tests, please also report that *and* how you tested the assumptions!!
- Report if there were missing data and how you dealt with them
- Precise description of the results
 - ‘6/8 participants rated condition X higher.’ instead of ‘Most participants liked condition X.’
 - ‘The mean of group X was 5.2 whereas the mean of group Y was more than one point lower (3.8).’ instead of ‘The mean of group X was higher than the mean of group Y.’
- Please use ‘significant’ only if you are talking about statistical significance (otherwise it’s confusing for a statistics geek like me)
- You should have an intuitive structure, e.g. with one subsection for each hypothesis

Do's (and Don'ts): Connection to Hypothesis

- Clear connection to the hypotheses (this applies to figures as well as results and interpretations)
- Explain why you mention some aspect or do some analysis
- Only report relevant information (e.g. don't show all responses to single items of a questionnaire if you are actually interested in the overall mean)
- The data analysis should be consistent with the hypotheses (e.g. don't measure and evaluate reaction time if your hypothesis is about perceived trustworthiness)
- Recommended: Structure your results according to the hypothesis, so that it is clear which result belongs to which hypothesis
- If you have additional exploratory analysis, please state them as such

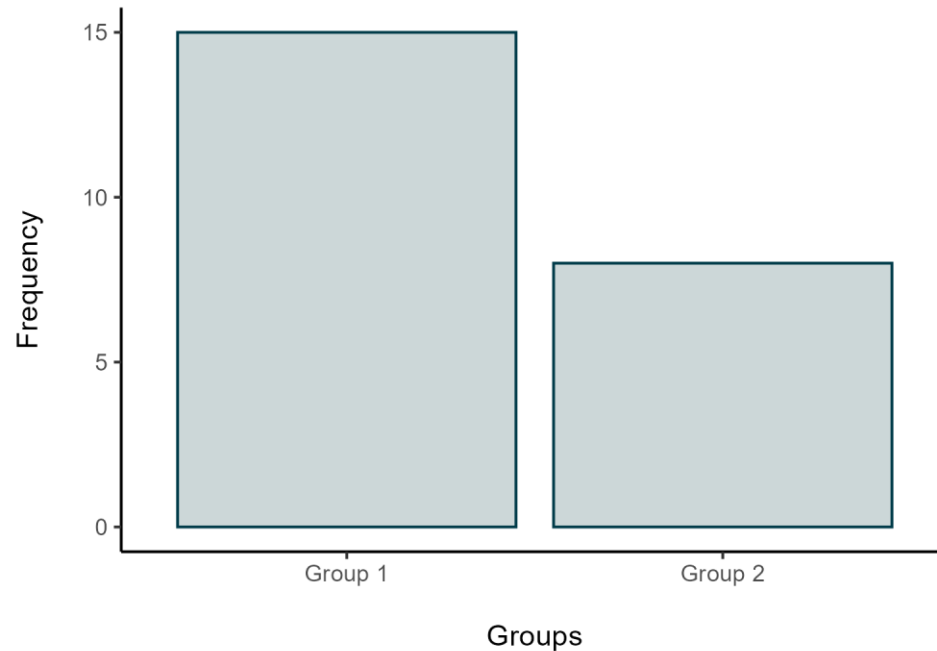
Do's (and Don'ts): Figures

- Comprehensible figures with relevant captions
 - Compact data presentation of most important results (not just everything)
 - Figure captions (should be understandable without reading the text)
 - Explanation and interpretation of the figures (what is presented and what should be seen, e.g. 'Half of the participants rated X higher than Y.')

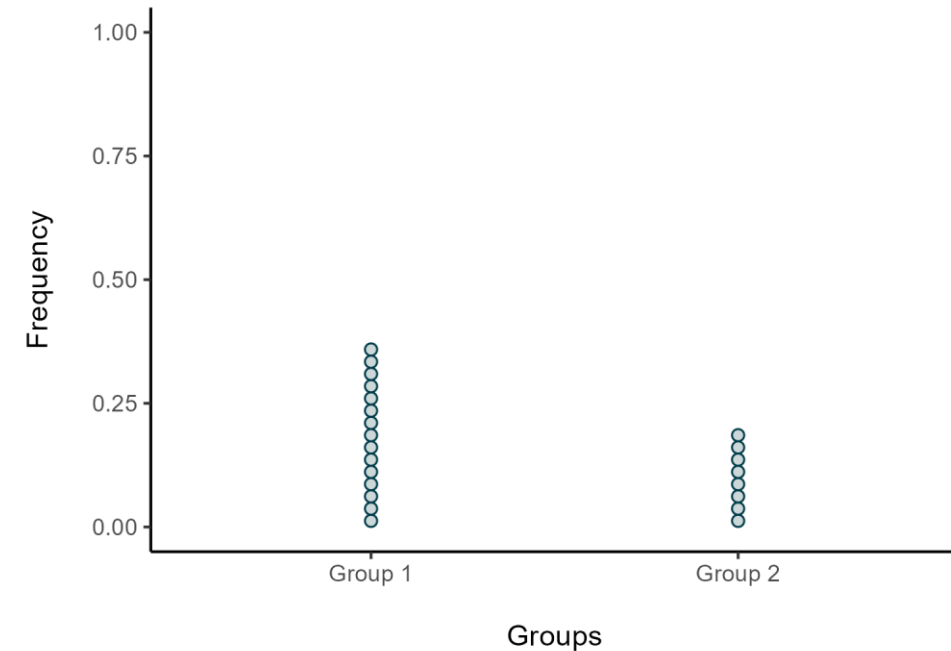
R script for plot generation is uploaded in Moodle

Plots: Categorical Variables

Bar chart

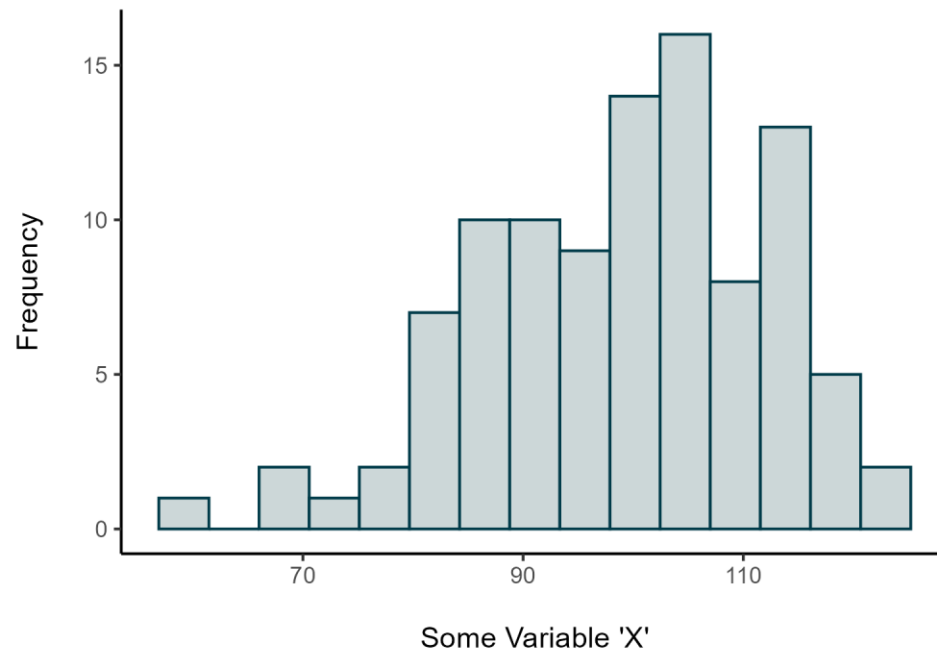


Dot plot

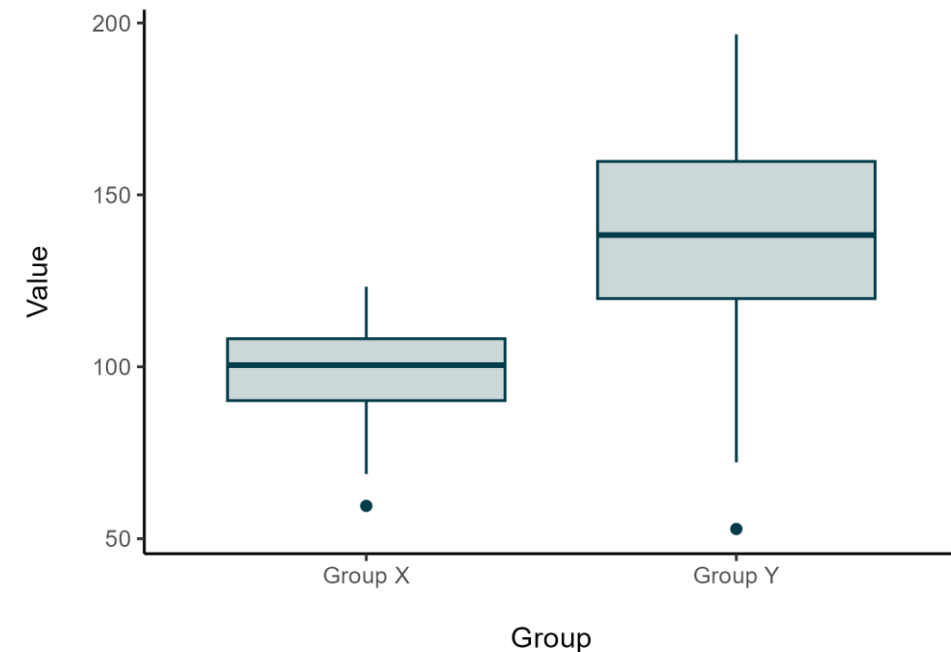


Plots: Continuous Variables

Histogram

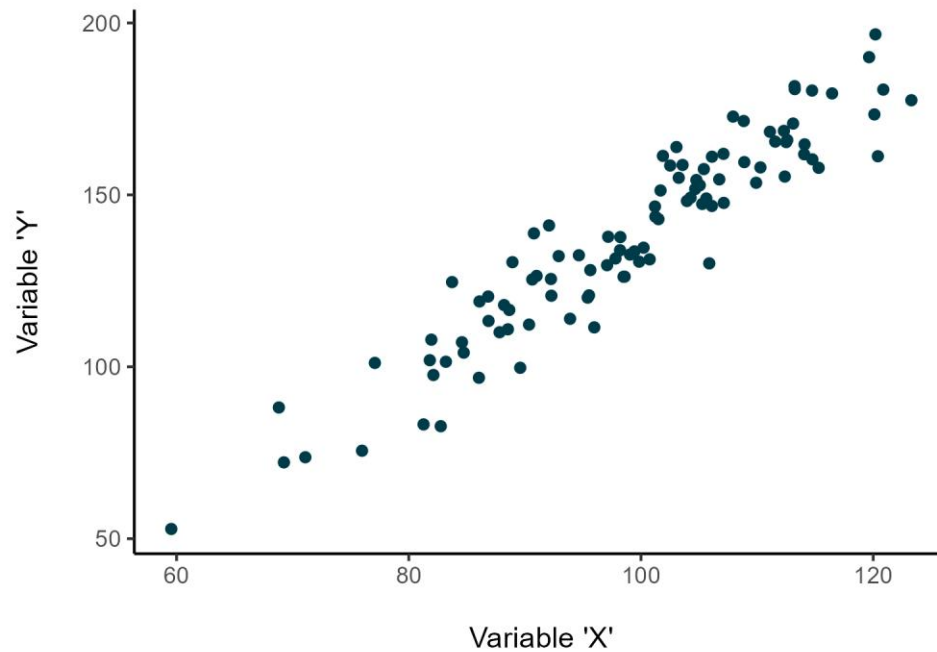


Boxplot

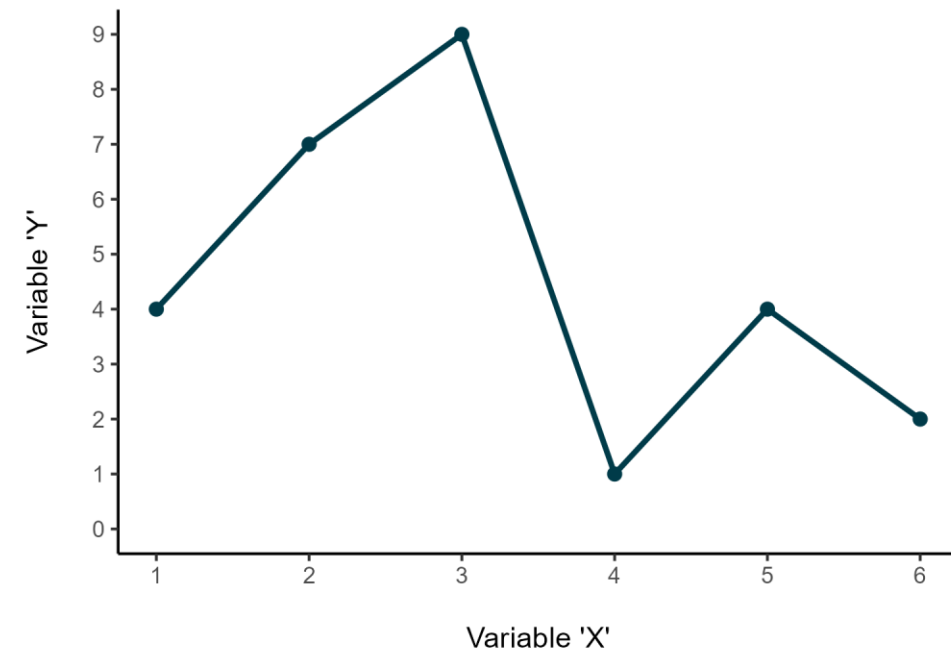


Plots: Relationships

Scatter plot

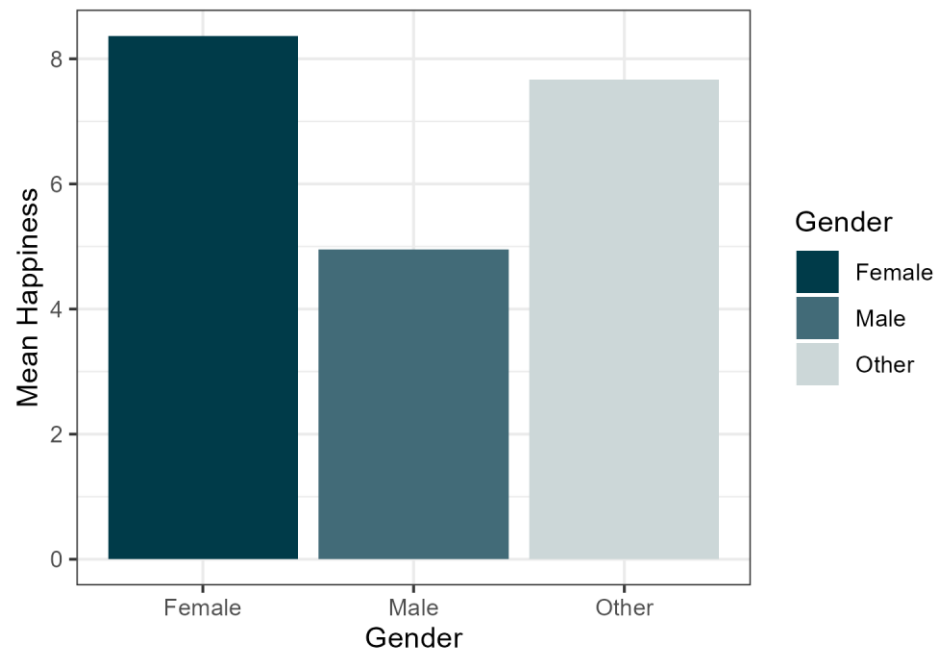


Line graph

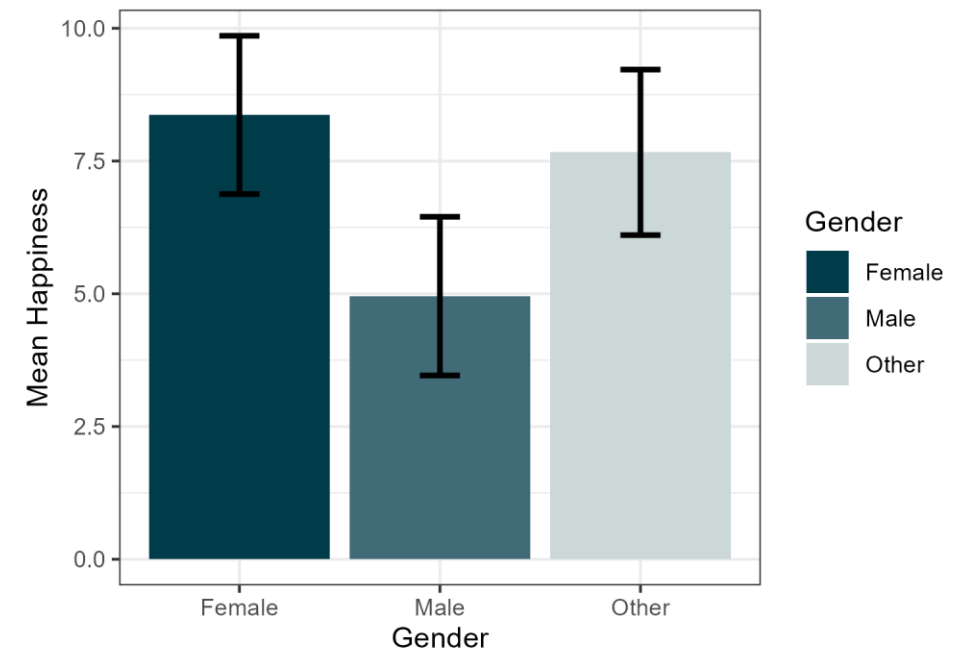


Plots: Error Bars

Mean



Mean + SD

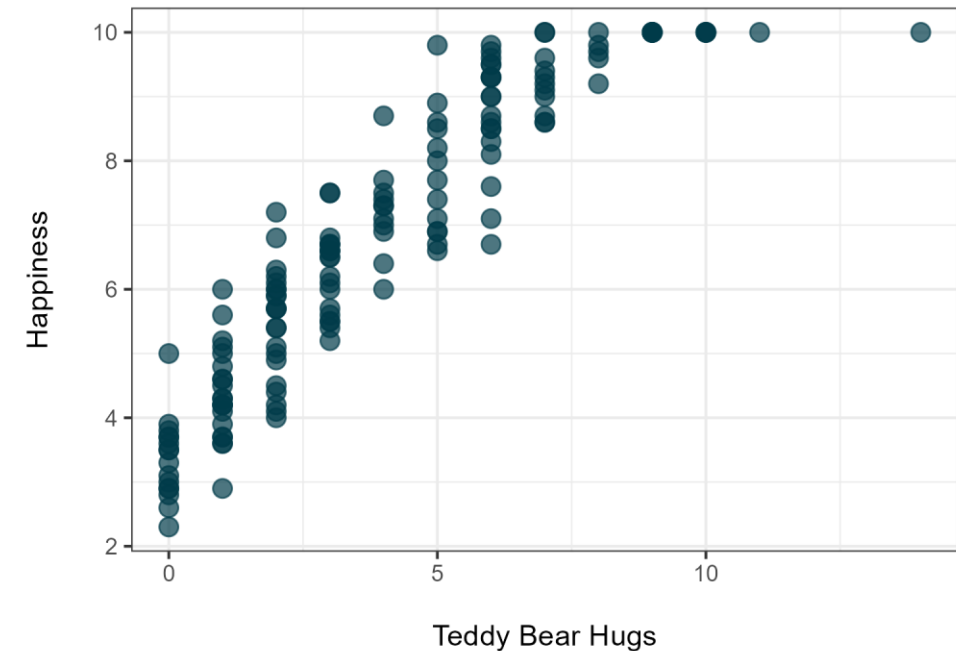
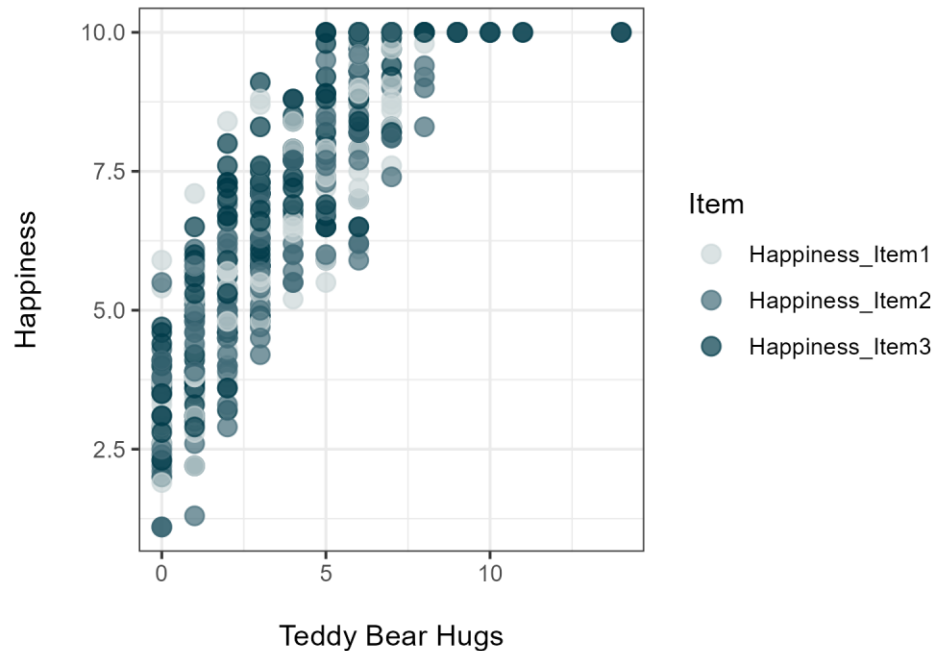


Plots: Level of Detail

Single Items

maybe too detailed?
really needed?

Average



Discussion

- Short summary of the main results again (this paragraph is often read first)
- Interpretation of the results in relation to the research question(s) and state if the results support the hypothesis or not
- Evaluation of the research question
- Links to literature: confirm, extend, or challenge prior work – also include some new literature!
- Possible explanations for unexpected findings
- Critical reflection of the study
- Limitations
- Suggestions for future research

- Advice: Structure into subsections (e.g. Summary, limitations, future work, etc.)

Conclusion

- Recap of key findings
- Possible implications for theory, practice, ...
- Outlook

Reference to Other Groups

- Please state similarities, overlaps, differences, and / or potential integrations into your idea.
- You can include this part at any place in the report that you find logical, e.g. after the theoretical background or in the discussion – that's up to you.
- You don't have to list all groups in all detail but should name them explicitly (not just "other groups did X"). You can for example group them according to their focus or go into more detail of some really interesting ones.
- You can compare the overall idea or implementation or other aspects.
- You should state clearly what is unique about your approach.

Title: Meaningful, Specific, and Concise

Example title	Informative Value
Project Report	Uninformative
Trustworthiness in Automated Systems	Overly broad and vague
How False Alarms Influence the Perceived Trustworthiness of a System	Focused but inconclusive
More False Alarms Do Not Affect the Perceived Trustworthiness of a System in a Multitasking Environment	Precise and informative
In a Multitasking Environment, Increasing the Number of False Alarms From One To Six Does Not Significantly Influence Users' Perceptions of the Trustworthiness and Reliability of the Automated Alert System	Too long and overly detailed
People Tolerate System Errors: False Alarms Do Not Reduce Perceived Trustworthiness	Catchy

⇒ Reflects the main topic and, ideally, the key finding or context

Abstract

- Brief summary (approximately 150 – 250 words)
- States purpose, method, main results, and conclusions
- Should be understandable without reading the full paper
- Keep it short and precise! Think about every information twice – is it really relevant enough to be mentioned in the abstract?

Evaluation Criteria in Moodle

Datei

Start

Einfügen

Seitenlayout

Formeln

Daten

Überprüfen

Ansicht

Hilfe

Acrobat

Teilen

Einfügen

Zwischenablage

Calibri

11

A⁺

A⁻

F

K

U

A

Standard

%

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0.00

→.0

Bedingte Formatierung

Als Tabelle formatieren

Zellenformatvorlagen

Einfügen

Löschen

Format

Sortieren und Filtern

Suchen und Auswählen

L32

Thank you for your attention!